Solid State Physics — Week 5 Assignment Questions & Answers

Question 41

The nearest neighbor distance in BCC structure is

(a) $\frac{a\sqrt{3}}{2}$

(b) $\frac{a\sqrt{3}}{\sqrt{2}}$

(c) $\frac{2a}{\sqrt{3}}$

(d) $\frac{3a}{\sqrt{2}}$

Accepted answer: (a)

Question 42

The fraction of volume unoccupied in the unit cell of a BCC lattice is

(a) $\frac{8-\pi\sqrt{3}}{8}$

(b) $\frac{\pi\sqrt{3}}{8}$

(c) $\frac{6-\pi\sqrt{2}}{6}$

(d) $\frac{\pi}{3\sqrt{2}}$

Accepted answer: (a)

Question 43

The mass of the each atom of a metal is w and its density is ρ. If the metal is FCC structure then the side 'a' of its conventional cubic cell is given by

(a) $\left(\frac{w}{\rho}\right)^{\frac{1}{3}}$

(b) $\left(\frac{2w}{\rho}\right)^{\frac{1}{3}}$

(c) $\left(\frac{4w}{\rho}\right)^{\frac{1}{3}}$

(d) $\left(\frac{6w}{\rho}\right)^{\frac{1}{3}}$

Accepted answer: (c)

Question 44

Find the Miller indices of the plane containing the three lattice points and their position vectors are $\vec{r}_1 = \vec{a} - \vec{b}$, $\vec{r}_2 = 2\vec{a} + \vec{c}$ and $\vec{r}_3 = 3\vec{b} + \vec{c}$ respectively. Consider $\vec{a}, \vec{b}$ and $\vec{c}$ are unit primitive vectors.

(a) (325)

(b) $(32\bar{5})$

(c) (121)

(d) $(1\bar{2}1)$

Accepted answer: (b)

Question 45

The smallest distance between any two atoms (nearest neighbor distance) in a FCC crystal is 0.2 nm. Find the volume of the unit cube.

(a) $0.0226\ (nm)^3$

(b) $0.0289\ (nm)^3$

(c) $0.0312\ (nm)^3$

(d) $0.0412\ (nm)^3$

Accepted answer: (a)

Question 46

Find the angle between [111] and [100] directions in cubic lattice.

(a) $22.82°$

(b) $43.47°$

(c) $54.73°$

(d) $82.12°$

Accepted answer: (c)

Question 47

Metallic copper forms cubic crystals and the lattice parameters is 0.36 nm. If the specific gravity of the copper is 9.04 and its atomic weight is 63.5, then find out its type of crystal structure.

(a) Simple cubic structure

(b) Body centred cubic structure

(c) Face centred cubic structure

(d) Hexagonal closed packed structure

Accepted answer: (c)

Question 48

An orthorhombic crystal has primitive translations a = 1.21 Å, b = 1.84 Å and c = 1.97 Å. If a plane with Miller indices $(23\bar{1})$ cuts an intercepts of 1.21 Å along the x-axis, find the lengths of intercepts along y and z axes.

(a) $1.23\ Å, -3.94\ Å$

(b) $1.43\ Å, -4.26\ Å$

(c) $1.57\ Å, -5.26\ Å$

(d) $1.76\ Å, -6.12\ Å$

Accepted answer: (a)

Question 49

The spacing between successive (100) planes in NaCl is 2.82 Å. X-rays incident on the surface of the crystal is found to give rise to first order Bragg reflection at glancing angle 8.8°. Calculate the wavelength of X-rays.

(a) 0.863 Å

(b) 1.12 Å

(c) 1.32 Å

(d) 1.54 Å

Accepted answer: (a)

Question 50

For a NaCl crystal, the cell-edge a = 0.563 nm. The smallest angle at which Bragg reflection can occur corresponds to a set of planes of indices.

(a) (100)

(b) (110)

(c) (111)

(d) (200)

Accepted answer: (a)